

# Fabrication of Graphene Microelectrodes for Biomedical Applications

Pranoti Kshirsagar <sup>a,b</sup>, Peter Jones <sup>a</sup>, Gerhard Heusel<sup>a</sup>, Thomas Chassé <sup>b,d</sup>, Monika Fleischer <sup>c,d</sup>, Claus J. Burkhardt <sup>a</sup>

<sup>a</sup>NMI Natural and Medical Sciences Institute at the University of Tübingen, 72770 Reutlingen (Germany)

<sup>b</sup>Center for Light-Matter-Interaction, Sensors and Analytics (LISA+), Eberhard Karls University Tübingen, 72076 Tübingen (Germany)

<sup>c</sup>Institute of Physical and Theoretical Chemistry, Eberhard Karls University Tübingen, 72076 Tübingen (Germany)

<sup>d</sup>Institute for Applied Physics, Eberhard Karls University Tübingen, 72076 Tübingen (Germany)

e-mail: pranoti.kshirsagar@nmi.de

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Microelectrode arrays (MEAs) are used for in-vitro recording of cellular activity. MEA devices consists of a base substrate such as float-glass, with micro-patterned conduction tracks and electrodes. The assembly is covered with an insulator where the electrodes (for contact with cell culture) and contact pads (electrical contact for recording) are opened. Electrogenic cells such as neurons or cardiomyocytes are cultured on the electrode array and their electrical activity is recorded. Conventional electrodes are made of opaque materials such as titanium nitride, iridium oxide, gold, CNT and PEDOT-CNT [1]. For simultaneous electrophysiology and visualization of the electrode-cell interface the electrode needs to transparent. As seen in figure 1(a), visualization of the cell-electrode interface is only possible from the bottom. Furthermore, transparent electrodes pave the way towards supporting research in fields such as optogenetics, calcium imaging and better understanding of neuronal networks. In 2014, a graphene MEA [2] was reported which kick started the use of this carbon form in the field of biotechnology. In this case the electrode diameter was 200  $\mu\text{m}$  which is too large if for good spatial resolution while recording cellular activity.

Here we present fabrication of 30  $\mu\text{m}$  diameter transparent graphene electrodes and their successful application with neurons, cardiac cells and retina slices.

Graphene was grown by chemical vapour deposition on copper foil with methane as precursor in presence of argon and hydrogen. Graphene was transferred to glass substrates with gold conduction lines which are terminated before the microelectrode. Graphene was structured by photolithography to obtain 30  $\mu\text{m}$  diameter electrodes. Figure 1(b) shows a scanning electrode micrograph of the electrode field with conduction lines terminated 5  $\mu\text{m}$  before the electrode. SU-8 was used as the insulator.

Graphene MEAs were validated with cultured neuronal and cardiac cells and retinal preparations. Figure 2 shows green fluorescent protein (GFP) labelled retinal ganglion cells in a mouse retina on the electrode array. A neuron in direct contact with a transparent graphene electrode is indicated. In case of opaque electrodes this optical access would be lost which proves the advantage of replacing conventional electrodes materials with graphene. Visualization in combination with recordings can provide better understanding of the neuronal network.

Figure 3 shows recordings from cultured rat cortical neurons. Here the signal to noise ratio was approximately 9.

In summary, reliable and reproducible transparent graphene microelectrodes were fabricated and their application was demonstrated in multiple neuronal and cardiac applications.

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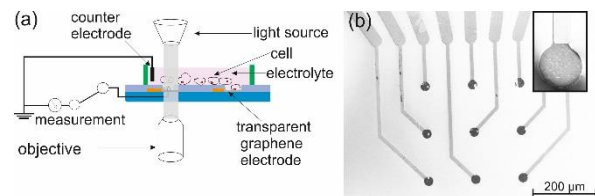


Figure 1: (a) Schematic showing side view of the MEA with cell culture on top and microscope objective at the bottom. (b) Scanning electron micrograph showing a top-view of an array of nine graphene electrodes. Inset shows one of the 30  $\mu\text{m}$  electrode.

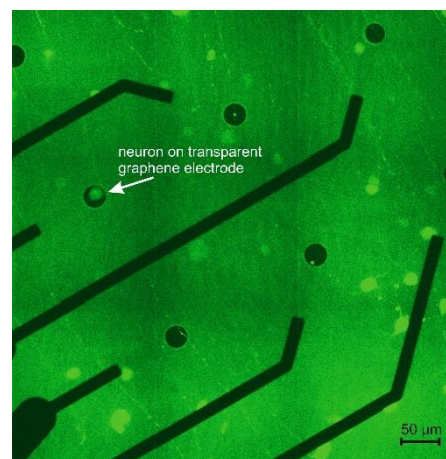


Figure 2: Confocal microscopy of mouse retina with green fluorescent protein (GFP) labelled retinal ganglion cells on graphene MEA. Also visible are the gold + graphene conduction lines (dark / transparent) and autofluorescence of the SU-8 insulator.

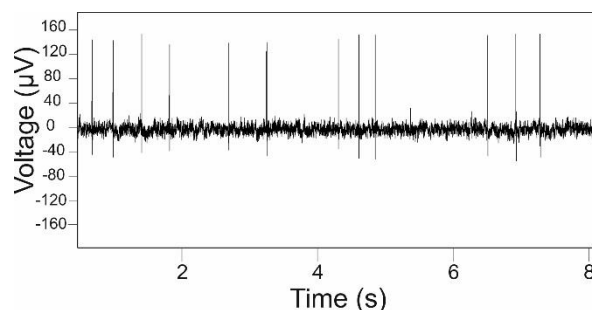


Figure 3: Action potentials recorded from rat cortical neurons cultured on graphene MEAs. Signals of up to 140  $\mu\text{V}$  were easily distinguished from the noise of  $\sim 15$   $\mu\text{V}$ .

[1] R. Gerwig, et. al., *Front. Neuroeng.*, 2014

[2] D.-W. Park et. al., *Nature communications*, 4, 5258, 2014